Notes: Stambaugh, Yu, and Yuan (JF, 2015)

Arbitrage Asymmetry and the Idiosyncratic Volatility Puzzle

Contents

1 Big picture

- **Question.** Why is the average return–idiosyncratic volatility relation negative, even though idiosyncratic volatility should be diversifiable in classical asset pricing theory?
- **Main explanation.** Idiosyncratic volatility is arbitrage risk. It allows mispricing to persist. Because buying is easier than shorting, arbitrage is asymmetric: underpricing is corrected more aggressively than overpricing.
- **Core mechanism.** High-IVOL overpriced stocks remain more overpriced because short-side arbitrage is limited. High-IVOL underpriced stocks are less severely underpriced because long-side arbitrage is easier.
- **Aggregate result.** The negative IVOL effect among overpriced stocks dominates the positive IVOL effect among underpriced stocks, producing the overall negative IVOL-return relation.
- **Mispricing measure.** The paper constructs a composite mispricing ranking from 11 return anomalies. Stocks ranked as most overpriced or most underpriced are compared across IVOL quintiles.
- **Main portfolio result.** Among the most overpriced stocks, high-IVOL stocks earn much lower abnormal returns than low-IVOL stocks. Among the most underpriced stocks, high-IVOL stocks earn higher abnormal returns, but the positive relation is weaker.
- **Short-sale constraint evidence.** The negative IVOL effect among overpriced stocks is stronger when stocks are harder to short, proxied by low institutional ownership and smaller market capitalization.
- **Sentiment evidence.** High investor sentiment strengthens the negative IVOL effect among overpriced stocks. Low sentiment strengthens the positive IVOL effect among underpriced stocks.
- **Economic takeaway.** The IVOL puzzle is not a stand-alone risk anomaly. It is a mispricing-and-limits-to-arbitrage phenomenon shaped by asymmetric arbitrage constraints.

2 Data and measurement

2.1 Sample and returns

- The sample covers U.S. common stocks from August 1965 to January 2011.
- Monthly stock returns are adjusted using the Fama–French three-factor benchmark.
- The benchmark-adjusted return is estimated from the regression

$$R_{i,t} = a + b\text{MKT}_t + c\text{SMB}_t + d\text{HML}_t + \epsilon_{i,t}. \tag{1}$$

- The abnormal return used in tables is the intercept-style benchmark-adjusted component, or the return residual relative to the three-factor model.

Interpretation: The factor adjustment ensures that the IVOL effects are not simply compensation for exposure to the market, size, or value factors.

2.2 Idiosyncratic volatility

- IVOL is measured following Ang et al. (2006).
- For each stock, daily returns over the most recent month are regressed on Fama–French factors.
- IVOL is the standard deviation of the regression residuals.
- Stocks are sorted into five IVOL groups each month.

Interpretation: IVOL is treated not as a priced risk factor, but as arbitrage risk: the risk that makes arbitrage capital reluctant to attack mispricing.

2.3 Composite mispricing measure

- The paper combines 11 anomaly variables that survive Fama–French adjustment.
- Each stock receives a ranking percentile for each anomaly.
- The composite mispricing score is the average of the 11 ranking percentiles.
- Stocks are divided into five mispricing groups: most overpriced, next 20%, middle, next 20%, and most underpriced.

Interpretation: The composite measure reduces dependence on any single anomaly and captures broad cross-sectional mispricing.

2.4 Institutional ownership and shorting ease

- Institutional ownership comes from Thomson Financial institutional holdings data.
- The high-IO group and low-IO group are based on the top and bottom 30% of residual institutional ownership after controlling for size and size squared.
- Low institutional ownership proxies for stocks that are harder to short.

Interpretation: If arbitrage asymmetry is central, the negative IVOL effect among overpriced stocks should be stronger where shorting is more difficult.

2.5 Investor sentiment

- The paper uses the Baker–Wurgler market-wide investor sentiment index.
- High-sentiment months are above-median sentiment months; low-sentiment months are below-median sentiment months.
- Sentiment is also used continuously in predictive regressions.

Interpretation: Sentiment shifts the market-wide direction of mispricing. High sentiment makes overpricing more likely, while low sentiment makes underpricing more likely.

3 Models and empirical specifications

3.1 Simple mechanism

Let M_i denote a stock's mispricing, where high values indicate overpricing and low values indicate underpricing. Let IVOL_i denote arbitrage risk.

The model's qualitative implications are:

$$\frac{\partial E[R_i]}{\partial \text{IVOL}_i} < 0 \quad \text{for overpriced stocks,} \tag{2}$$

$$\frac{\partial E[R_i]}{\partial \text{IVOL}_i} > 0 \quad \text{for underpriced stocks.} \tag{3}$$

Arbitrage asymmetry implies

$$\left|\frac{\partial E[R_i]}{\partial \text{IVOL}_i}\right|_{\text{overpriced}} > \left|\frac{\partial E[R_i]}{\partial \text{IVOL}_i}\right|_{\text{underpriced}}. \tag{4}$$

Interpretation: The negative relation among overpriced stocks is stronger because shorting constraints restrict arbitrage more than buying constraints.

3.2 Double-sort portfolio specification

Each month, stocks are independently sorted into:

- five mispricing groups based on the composite anomaly score;
- five IVOL groups based on recent idiosyncratic volatility.

The key portfolio spread is:

$$IVOLSpread = R_{High\ IVOL} - R_{Low\ IVOL}. \tag{5}$$

Interpretation: A negative spread among overpriced stocks and a positive spread among underpriced stocks is the central empirical prediction.

3.3 Function of mispricing

The paper estimates a smooth relation between IVOL effects and mispricing:

$$f(M) = \text{effect of standardized IVOL on abnormal return for stocks with mispricing rank } M. \tag{6}$$

Interpretation: The function $f(M)$ is expected to slope downward: IVOL effects are positive for underpricing and negative for overpricing, with stronger magnitudes at the extremes.

3.4 Sentiment predictive regression

For each mispricing category and IVOL leg, the paper estimates:

$$R_{i,t} = a + bS_{t-1} + c\text{MKT}_t + d\text{SMB}_t + e\text{HML}_t + u_t, \tag{7}$$

where S_{t-1} is the Baker–Wurgler sentiment index at the end of the previous month.

With macro controls, the specification becomes:

$$R_{i,t} = a + b\widetilde{S}_{t-1} + c\text{MKT}_t + d\text{SMB}_t + e\text{HML}_t + \sum_{j=1}^{6} m_j X_{j,t-1} + u_t. \tag{8}$$

Interpretation: A negative coefficient on sentiment for the high-minus-low IVOL spread means that the negative IVOL effect strengthens when sentiment is high.

4 Identification and empirical design

4.1 What identifies the explanation

- The overall negative IVOL relation is decomposed by the direction of mispricing.
- The paper tests whether high-IVOL stocks are worse performers only when they are overpriced.
- It tests whether the negative IVOL effect is stronger when shorting is harder.
- It tests whether the sign and strength of the IVOL effect vary with investor sentiment.

Interpretation: The identification comes from matching several distinct predictions of the arbitrage-asymmetry theory, not from a single return sort.

4.2 Why the mispricing split matters

- If IVOL were simply a risk factor, its return relation should not reverse sign across overpricing and underpricing groups.
- If IVOL were simply a lottery-demand proxy, the direction should not line up so cleanly with anomaly-based mispricing.
- The sign flip across mispricing groups is therefore crucial.

Interpretation: The mispricing split converts the IVOL puzzle into a limits-to-arbitrage prediction.

4.3 Why institutional ownership matters

- Low institutional ownership proxies for limited lendable supply and harder shorting.
- The model predicts stronger negative IVOL effects among overpriced low-IO stocks.
- This is exactly the pattern in Table III.

Interpretation: The shorting-ease evidence is the stock-level arbitrage-asymmetry test.

4.4 Why sentiment matters

- Sentiment proxies for market-wide overpricing or underpricing pressure.
- High sentiment should strengthen the negative IVOL effect among overpriced stocks.
- Low sentiment should strengthen the positive IVOL effect among underpriced stocks.
- Because arbitrage asymmetry favors long-side arbitrage, the overpricing/sentiment effect should be stronger.

Interpretation: Sentiment provides time-series variation in the direction and intensity of mispricing.

4.5 Main limitations

- The mispricing ranking is a proxy, not direct observation of true mispricing.
- Institutional ownership is a proxy for shorting ease, not a complete short-sale constraint measure.
- Sentiment may be correlated with macroeconomic variables, which is why the paper adds macro controls.
- Return predictability is always vulnerable to data-mining concerns, although the use of broad anomaly composites reduces reliance on any one signal.

Interpretation: The evidence is persuasive because multiple independent implications point in the same direction.

5 Tables and figures

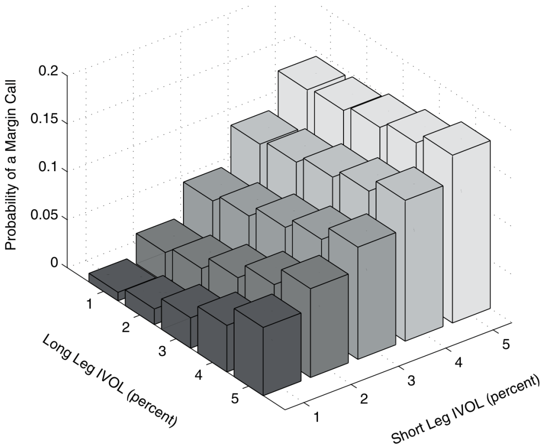
The tables and figures below are directly cropped from the paper and grouped compactly.

5.1 Figure 1 and Table I: arbitrage risk and the double-sort sample

Read:

- Figure 1 shows that margin-call probability is more sensitive to short-leg IVOL than to long-leg IVOL.
- Table I shows that IVOL rises monotonically across IVOL-sorted portfolios.
- Overpriced portfolios contain more high-IVOL stocks; underpriced portfolios contain more low-IVOL stocks.

Economic message: Short-side arbitrage is more fragile when the short leg is volatile, and the sample already shows a tendency for high-IVOL stocks to be more overpriced.



igure 1. IVOL and the probability of a margin call. The figure plots the proba-
ility of a long-short strategy hitting a 25% maintenance margin level within the next
2 months when the current margin level is 35%. The current long and short positions are of
qual size and have monthly IVOL values between 1% and 5%. The long (short) leg has a monthly
lpha of 0.5% (−0.5%), and both legs have betas equal to one. The market portfolio's monthly

Figure 1: IVOL and margin-call probability

Table I

Individual Stock IVOL and Number of Stocks in the Double-Sorted Portfolios

Panel A reports the typical individual stock IVOL within each portfolio, first computing the median IVOL each month and then averaging the medians across months. Panel B reports the average number of stocks in each portfolio. The 25 portfolios are formed by independently sorting on IVOL and the mispricing measure. The latter is the average of the ranking percentiles produced by 11 anomaly variables. We compute IVOL, following Ang et al. (2006), as the standard deviation of the most recent month's daily benchmark-adjusted returns, with the latter equal to the residuals in a regression of each stock's daily return on the three factors defined by Fama and French (1993): MKT, SMB, and HML. The sample period is from August 1965 to January 2011.

	Highest IVOL	Next 20%	Next 20%	Next 20%	Lowest IVOL	All Stocks
			Panel A: IVOL			
Most Overpriced	3.36	1.47	0.82	0.46	0.20	1.29
Next 20%	3.22	1.45	0.81	0.45	0.19	0.88
Next 20%	3.15	1.44	0.81	0.45	0.19	0.75
Next 20%	3.11	1.43	0.80	0.44	0.19	0.68
Most Underpriced	3.06	1.42	0.80	0.44	0.19	0.63
All Stocks	3.21	1.44	0.81	0.45	0.19	0.81
			Panel B: Number of Stocks			
Most Overpriced	196	148	115	90	73	622
Next 20%	131	132	128	120	111	623
Next 20%	110	121	127	131	133	623
Next 20%	98	114	127	138	145	623
Most Underpriced	88	107	125	144	159	623
All Stocks	622	623	623	623	622	3,113

Table I: IVOL and number of stocks

5.2 Table II and Figure 2: IVOL effects by mispricing direction

Read:

- Table II reports the central double-sort result.
- Among the most overpriced stocks, the high-minus-low IVOL return is −1.50% per month.
- Among the most underpriced stocks, the high-minus-low IVOL return is +0.41% per month.
- Across all stocks, the high-minus-low IVOL return is −0.78% per month.
- Figure 2 visualizes the same sign reversal: negative IVOL effects among overpriced stocks and positive IVOL effects among underpriced stocks.

Economic message: The IVOL puzzle is strongest where the theory predicts: high-IVOL overpriced stocks underperform sharply, while high-IVOL underpriced stocks outperform more modestly.

Table II

Idiosyncratic Volatility Effects in Underpriced versus Overpriced Stocks

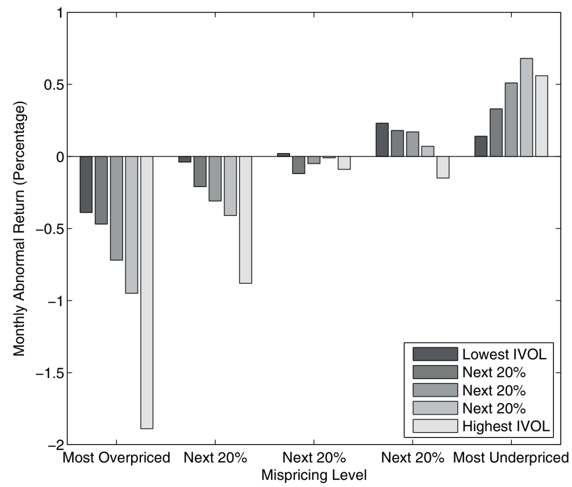
This table reports average benchmark-adjusted returns for portfolios formed by sorting stocks independently on IVOL and the mispricing measure. The mispricing measure is the average of the ranking percentiles produced by 11 anomaly variables. Also reported are results based on sorting by IVOL within the entire stock universe. Benchmark-adjusted returns are calculated as a in the regression

$$R_{i,t} = a + bMKT_t + cSMB_t + dHML_t + \epsilon_{i,t},$$

where $R_{i,t}$ is the excess percent return in month t. The sample period is from August 1965 to January 2011. All t-statistics (in parentheses) are based on the heteroskedasticity-consistent standard errors of White (1980).

	Highest IVOL	Next 20%	Next 20%	Next 20%	Lowest IVOL	Highest −Lowest	All Stocks
Most Overpriced	−1.89	−0.95	−0.72	−0.47	−0.39	−1.50	−0.81
(Top 20%)	(−12.05)	(−7.39)	(−4.90)	(−3.62)	(−3.04)	(−7.36)	(−8.14)
Next 20%	−0.88	−0.41	−0.31	−0.21	−0.04	−0.84	−0.23
	(−5.86)	(−3.36)	(−3.00)	(−2.08)	(−0.44)	(−4.41)	(−3.88)
Next 20%	−0.09	−0.01	−0.05	−0.12	0.02	−0.10	−0.07
	(−0.53)	(−0.09)	(−0.48)	(−1.29)	(0.18)	(−0.53)	(−1.47)
Next 20%	−0.15	0.07	0.17	0.18	0.23	−0.38	0.18
	(−0.80)	(0.63)	(1.87)	(2.33)	(3.22)	(−1.78)	(4.45)
Most Underpriced	0.56	0.68	0.51	0.33	0.14	0.41	0.28
(Bottom 20%)	(3.27)	(4.91)	(5.02)	(4.10)	(2.04)	(2.16)	(5.67)
Most Overpriced −	−.44	−1.63	−1.23	−0.81	−0.53	−1.91	−1.09
Most Underpriced	(−11.07)	(−8.65)	(−6.43)	(−5.02)	(−3.43)	(−7.62)	(−8.05)
All Stocks	−0.69	−0.12	−0.00	0.05	0.08	−0.78	
	(−6.09)	(−1.56)	(−0.01)	(1.07)	(1.86)	(−5.50)	

Table II: IVOL effects in overpriced versus underpriced stocks



ure 2. Monthly abnormal returns of portfolios ranked by mispricing level and IVOL
figure plots the average monthly abnormal return on portfolios formed in a 5 × 5 sort tha

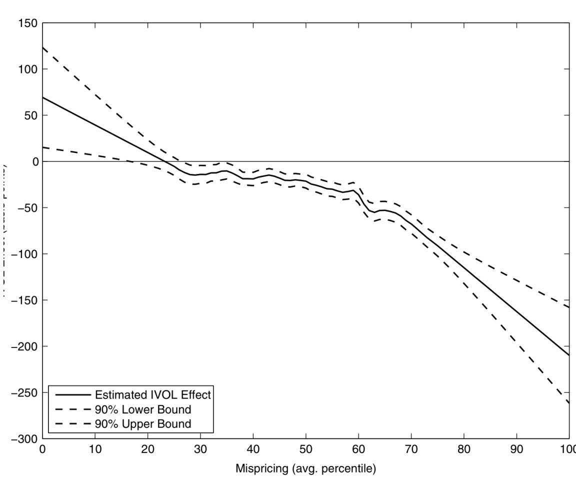
Figure 2: abnormal returns across IVOL and mispricing groups

5.3 Figure 3 and Table III: estimated IVOL effect and shorting ease

Read:

- Figure 3 plots $f(M)$, the estimated effect of standardized IVOL by mispricing rank.
- The effect becomes increasingly negative as stocks move toward the overpriced tail.
- Table III shows that the negative IVOL effect is stronger among low-institutional-ownership stocks.
- For the most overpriced stocks, the low-IO high-minus-low IVOL spread is more negative than the high-IO spread.

Economic message: The evidence directly supports stock-level arbitrage asymmetry: harder-to-short overpriced stocks show the strongest negative IVOL effect.



gure 3. Estimated IVOL effects. The figure plots estimates of $f(M)$, which is the effect standardized IVOL on the abnormal monthly return for a stock whose mispricing ranking

Figure 3: estimated IVOL effects

Idiosyncratic Volatility Effects in Subsamples of High versus Low Institutional Ownership

This table reports average benchmark-adjusted returns for portfolios constructed by sorting independently on IVOL and IO within each quintile of the mispricing measure. The high-IO (low-IO) subsample consists of the top (bottom) 30% of stocks sorted on size-adjusted IO, computed following Nagel (2005): each quarter, we regress the logit of the IO percentage on log(size) and the square of log(size) and take the regression residual as size-adjusted IO. The data on institutional holdings come from the Thomson Financial Institutional Holdings database. The mispricing quintiles are determined by sorting on the average ranking percentiles produced by 11 anomaly variables. Also reported are results based on sorting by IVOL and IO within the entire stock universe. The benchmark-adjusted returns are estimates of a in the regression

$$R_{i,t} = a + bMKT_t + cSMB_t + dHML_t + \epsilon_{i,t},$$

where $R_{i,t}$ is the excess percent return in month t. The sample period is from April 1980 to January 2011. All t-statistics (in parentheses) are based on the heteroskedasticity-consistent standard errors of White (1980).

	High-IO Sample			Low-IO Sample			High-IO Sample − Low-IO Sample		
	Highest IVOL	Lowest IVOL	Highest −Lowest	Highest IVOL	Lowest IVOL	Highest −Lowest	Highest IVOL	Lowest IVOL	Highest −Lowest
Most Overpriced	−2.65	−0.61	−2.03	−3.11	−0.16	−2.95	0.47	−0.45	0.92
(Top 20%)	(−8.41)	(−3.01)	(−5.42)	(−9.31)	(−0.80)	(−7.48)	(1.14)	(−1.87)	(1.94)
Next 20%	−0.95	−0.09	−0.86	−1.25	0.14	−1.39	0.30	−0.23	0.52
	(−3.44)	(−0.48)	(−2.70)	(−4.56)	(0.84)	(−4.34)	(0.82)	(−1.08)	(1.29)
Next 20%	−0.61	0.14	−0.75	−0.23	0.31	−0.54	−0.38	−0.17	−0.21
	(−2.24)	(0.87)	(−2.29)	(−0.77)	(2.16)	(−1.52)	(−0.93)	(−0.82)	(−0.44)
Next 20%	−0.01	0.30	−0.31	−0.16	0.10	−0.26	0.15	0.20	−0.05
	(−0.04)	(1.94)	(−0.95)	(−0.59)	(0.74)	(−0.83)	(0.43)	(0.99)	(−0.11)
Most Underpriced	0.59	0.19	0.40	0.79	0.17	0.62	−0.20	0.01	−0.22
(Bottom 20%)	(2.18)	(1.36)	(1.29)	(3.34)	(1.28)	(2.14)	(−0.61)	(0.08)	(−0.57)
Most Overpriced −	−3.23	−0.80	−2.43	−3.90	−0.33	−3.57	0.67	−0.47	1.14
Most Underpriced	(−8.08)	(−3.07)	(−5.00)	(−10.15)	(−1.26)	(−7.89)	(1.36)	(−1.51)	(1.89)
All Stocks	−0.65	0.20	−0.86	−1.06	0.17	−1.23	0.41	0.03	0.38
	(−3.66)	(2.46)	(−4.01)	(−5.06)	(1.94)	(−5.13)	(1.72)	(0.29)	(1.49)

Nagel (2005) observes that, within the overall stock universe, the negative

Table III: IVOL effects by institutional ownership

5.4 Table IV, Figure 4, and Figure 5: investor sentiment

Read:

- Table IV shows that the negative IVOL effect among overpriced stocks is stronger after high sentiment.
- The positive IVOL effect among underpriced stocks is stronger after low sentiment.
- Figure 4 compares high- and low-sentiment periods for the most overpriced and most underpriced portfolios.
- Figure 5 shows that the estimated IVOL effect following high sentiment is more negative across the overpricing region.

Economic message: Sentiment shifts the market-wide direction of mispricing. The IVOL effect reacts exactly as the theory predicts.

Table IV

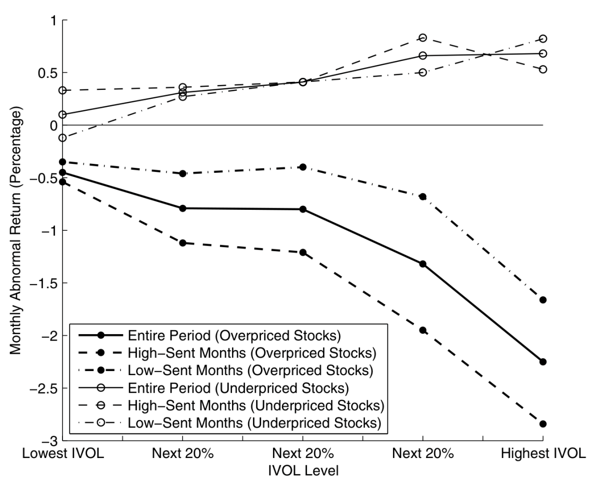
Idiosyncratic Volatility Effects in High-Sentiment versus Low-Sentiment Periods

This table reports average benchmark-adjusted returns for portfolios containing stocks with either the highest (top 20%) or lowest (bottom 20%) IVOL. The sort on IVOL is performed for stocks within a given range of over-/underpricing, as determined by an average of the ranking percentiles produced by 11 anomaly variables. Also reported are results based on sorting by IVOL within the entire stock universe. The benchmark-adjusted returns in high- and low-sentiment periods are estimates of a_H and a_L in the regression

$$R_{i,t} = a_H d_{H,t} + a_L d_{L,t} + bMKT_t + cSMB_t + dHML_t + \epsilon_{i,t},$$

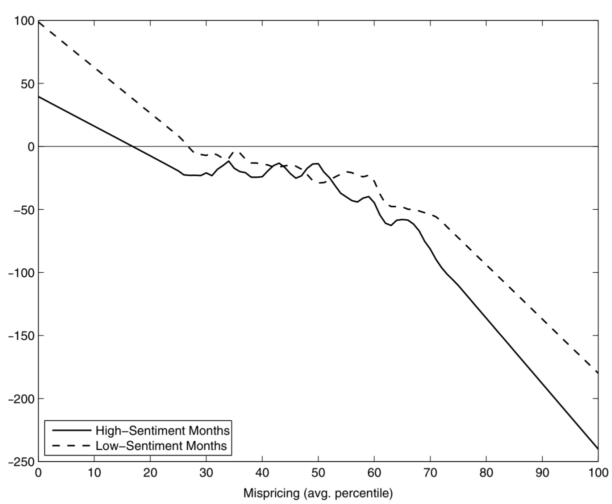
where $d_{H,t}$ and $d_{L,t}$ are dummy variables indicating high- and low-sentiment periods, and $R_{i,t}$ is the excess percent return in month t. The sample period is from August 1965 to January 2011. All t-statistics (in parentheses) are based on the heteroskedasticity-consistent standard errors of White (1980).

	High-Sentiment Periods			Low-Sentiment Periods			High-Sentiment Periods − Low-Sentiment Periods		
	Highest IVOL	Lowest IVOL	Highest −Lowest	Highest IVOL	Lowest IVOL	Highest −Lowest	Highest IVOL	Lowest IVOL	Highest −Lowest
Most Overpriced	2.84	−0.54	−2.30	−1.66	−0.36	−1.30	−1.18	−0.18	−1.00
(Top 20%)	(−9.57)	(−3.13)	(−6.79)	(−6.91)	(−2.55)	(−4.75)	(−3.06)	(−0.86)	(−2.29)
Next 20%	−1.24	−0.01	−1.23	−0.60	−0.16	−0.44	−0.64	0.15	−0.79
	(−5.28)	(−0.04)	(−4.31)	(−2.77)	(−1.26)	(−1.71)	(−2.02)	(0.82)	(−2.07)
Next 20%	−0.17	0.31	−0.48	−0.10	−0.22	0.13	−0.07	0.53	−0.60
	(−0.72)	(2.34)	(−1.75)	(−0.54)	(−1.92)	(0.52)	(−0.25)	(3.09)	(−1.68)
Next 20%	−0.10	0.19	−0.29	−0.04	0.11	−0.16	−0.06	0.08	−0.14
	(−0.35)	(1.44)	(−0.84)	(−0.23)	(1.29)	(−0.75)	(−0.18)	(0.49)	(−0.34)
Most Underpriced	0.54	0.33	0.21	0.82	−0.12	0.94	−0.28	0.45	−0.73
(Bottom 20%)	(2.43)	(2.77)	(0.77)	(4.05)	(−1.21)	(4.16)	(−0.93)	(2.85)	(−2.03)
Most Overpriced −	−3.38	−0.87	−2.51	−2.48	−0.24	−2.24	−0.90	−0.63	−0.27
Most Underpriced	(−9.36)	(−4.02)	(−6.48)	(−7.82)	(−1.22)	(−6.60)	(−1.85)	(−2.23)	(−0.53)
All Stocks	−1.06	0.26	−1.32	−0.33	−0.10	−0.23	−0.72	0.36	−1.09
	(−5.75)	(3.81)	(−5.88)	(−2.45)	(−1.87)	(−1.35)	(−3.16)	(4.16)	(−3.82)



ure 4. IVOL effects and investor sentiment. The figure plots the average monthly abnor
l return on portfolios formed in a 5 × 5 sort that ranks first by mispricing level and then by IVOL
ults are displayed for the five portfolios in the most underpriced quintile and the five portfolio

Figure 4: IVOL effects and investor sentiment



ıre 5. Estimated IVOL effects following high and low sentiment. The figure plots es-
tes of $f(M)$, which is the effect of standardized IVOL on the abnormal monthly return for

Figure 5: estimated IVOL effects by sentiment

5.5 Table V and Table VI: predictive sentiment regressions

Read:

- Table V regresses high-IVOL, low-IVOL, and high-minus-low IVOL returns on investor sentiment.
- Sentiment has a negative coefficient for high-minus-low IVOL spreads, especially among overpriced stocks.
- Table VI adds macroeconomic controls and orthogonalizes sentiment with respect to macro variables.
- The main sentiment results remain after adding controls.

Economic message: The sentiment evidence is not just a binary split. Continuous predictive regressions confirm that high sentiment strengthens the negative IVOL effect.

Table V
Idiosyncratic Volatility Effects and Investor Sentiment: Predictive Regressions

This table reports estimates of b in the regression

$$R_{i,t} = a + bS_{t-1} + cMKT_t + dSMB_t + eHML_t + u_t,$$

where $R_{i,t}$ is the excess percent return in month t, and S_t is the level of the investor sentiment index of Baker and Wurgler (2006). The sort on IVOL is performed for stocks within a given range of over-/underpricing, as determined by an average of the ranking percentiles produced by 11 anomaly variables. Also reported are results based on sorting by IVOL within the entire stock universe. The sample period is from August 1965 to January 2011. All t-statistics are based on the heteroskedasticity-consistent standard errors of White (1980).

	Highest IVOL		Lowest IVOL		Highest − Lowest	
	$\hat{b}$	t-Stat.	$\hat{b}$	t-Stat.	$\hat{b}$	t-Stat.
Most Overpriced (Top 20%)	−0.78	−3.74	0.01	0.08	−0.79	−3.49
Next 20%	−0.40	−2.50	0.09	0.97	−0.48	−2.50
Next 20%	−0.10	−0.74	0.30	3.20	−0.40	−2.18
Next 20%	−0.13	−0.81	0.05	0.60	−0.18	−0.93
Most Underpriced (Bottom 20%)	−0.12	−0.92	0.16	1.81	−0.28	−1.80
Most Overpriced − Most Underpriced	−0.66	−2.76	−0.15	−1.12	−0.50	−2.20
All Stocks	−0.48	−3.92	0.18	3.77	−0.66	−4.25

Table V: predictive sentiment regressions

Table VI
Idiosyncratic Volatility Effects and Investor Sentiment: Predictive Regressions with Macroeconomic Controls

This table reports estimates of b in the regressions

$$R_{i,t} = a + b\tilde{S}_{t-1} + cMKT_t + dSMB_t + eHML_t + u_t \quad \text{(Panel A)}$$

$$R_{i,t} = a + b\tilde{S}_{t-1} + cMKT_t + dSMB_t + eHML_t + \sum_{j=1}^{6} m_j X_{j,t-1} + u_t \quad \text{(Panel B)}.$$

where $R_{i,t}$ is the excess percent return in month t, $\tilde{S}_t$ is the level of the investor sentiment index of Baker and Wurgler (2006) that is orthogonalized with respect to six macrovariables, and $X_{1,t}, \cdots, X_{6,t}$ are the term premium, default premium, interest rate, inflation rate, surplus ratio, and consumption-wealth ratio. The sort on IVOL is performed for stocks within a given range of over-/underpricing, as determined by an average of the ranking percentiles produced by 11 anomaly variables. Also reported are results based on sorting by IVOL within the entire stock universe. The sample period is from August 1965 to January 2011. All t-statistics are based on the heteroskedasticity-consistent standard errors of White (1980).

	Highest IVOL		Lowest IVOL		Highest − Lowest	
	$\hat{b}$	t-Stat.	$\hat{b}$	t-Stat.	$\hat{b}$	t-Stat.
Panel A: $R_{i,t} = a + b\tilde{S}_{t-1} + cMKT_t + dSMB_t + eHML_t + u_t$						
Most Overpriced (Top 20%)	−0.74	−3.56	0.03	0.28	−0.76	−3.42
Next 20%	−0.45	−2.89	0.08	0.92	−0.53	−2.81
Next 20%	−0.17	−1.24	0.29	3.10	−0.46	−2.49
Next 20%	−0.17	−1.12	0.04	0.52	−0.22	−1.14
Most Underpriced (Bottom 20%)	−0.20	−1.54	0.15	1.63	−0.35	−2.22
Most Overpriced − Most Underpriced	−0.54	−2.28	−0.12	−0.87	−0.42	−1.88
All Stocks	−0.52	−4.31	0.17	3.53	−0.69	−4.50
Panel B: $R_{i,t} = a + b\tilde{S}_{t-1} + cMKT_t + dSMB_t + eHML_t + \sum_{j=1}^{6} m_j X_{j,t-1} + u_t$						
Most Overpriced (Top 20%)	−0.64	−2.68	−0.08	−0.67	−0.56	−2.15
Next 20%	−0.46	−2.52	0.04	0.41	−0.50	−2.29
Next 20%	−0.10	−0.65	0.25	2.56	−0.35	−1.73
Next 20%	−0.09	−0.49	0.09	0.97	−0.17	−0.83
Most Underpriced (Bottom 20%)	−0.18	−1.21	0.07	0.70	−0.24	−1.42
Most Overpriced − Most Underpriced	−0.46	−1.75	−0.14	−0.94	−0.32	−1.25
All Stocks	−0.50	−3.58	0.15	2.84	−0.65	−3.69

Table VI: macroeconomic controls

5.6 Table VII and Table VIII: excluding smaller firms

Read:

- Table VII repeats the main double-sort analysis after excluding the smallest 20%, 40%, 60%, and 80% of firms by market capitalization.
- The negative IVOL effect among overpriced stocks remains economically large after small firms are excluded.
- Table VIII repeats the sentiment regression after excluding small firms.
- The sentiment-related negative IVOL effect remains significant across market-cap thresholds.

Economic message: The IVOL puzzle is not only a microcap phenomenon. Size and short-sale constraints matter, but the arbitrage-asymmetry mechanism survives exclusions of smaller firms.

Table VII

Idiosyncratic Volatility Effects in Underpriced versus Overpriced Stocks under Various Thresholds for Market Capitalization

This table reports average benchmark-adjusted returns for portfolios formed by sorting stocks independently on IVOL and the mispricing measure. In each panel, the universe of stocks being sorted consists of all those with market capitalization exceeding a given percentile. The mispricing measure is determined by an average of the ranking percentiles produced by 11 anomaly variables. Also reported are results based on sorting by IVOL within the entire universe. Benchmark-adjusted returns are calculated as a in the regression

$$R_{i,t} = a + bMKT_t + cSMB_t + dHML_t + \epsilon_{i,t},$$

where $R_{i,t}$ is the excess percent return in month t. The sample period is from August 1965 to January 2011. All t-statistics (in parentheses) are based on the heteroskedasticity-consistent standard errors of White (1980). In Panels A, B, C, and D, the smallest 20%, 40%, 60%, and 80% of the firms are omitted from the portfolio formation, respectively.

	Highest IVOL	Next 20%	Next 20%	Next 20%	Lowest IVOL	Highest −Lowest	All Stocks
Panel A: 20% Smallest Stocks Omitted							
Most Overpriced	−1.66	−0.85	−0.56	−0.54	−0.35	−1.31	−0.79
(Top 20%)	(−10.74)	(−6.01)	(−4.06)	(−4.05)	(−2.76)	(−6.56)	(−7.88)
Next 20%	−0.86	−0.39	−0.31	−0.24	−0.04	−0.82	−0.25
	(−5.75)	(−3.28)	(−3.09)	(−2.40)	(−0.45)	(−4.42)	(−4.30)
Next 20%	−0.12	0.09	0.01	−0.12	0.03	−0.15	−0.04
	(−0.77)	(0.80)	(0.13)	(−1.27)	(0.40)	(−0.80)	(−0.92)
Next 20%	−0.08	0.16	0.19	0.11	0.21	−0.29	0.16
	(−0.43)	(1.45)	(2.11)	(1.47)	(2.91)	(−1.42)	(3.72)
Most Underpriced	0.46	0.73	0.47	0.31	0.15	0.31	0.28
(Bottom 20%)	(2.83)	(5.66)	(4.72)	(3.78)	(2.25)	(1.64)	(5.68)
Most Overpriced −	−2.12	−1.58	−1.03	−0.85	−0.50	−1.62	−1.07
Most Underpriced	(−9.84)	(−8.22)	(−5.69)	(−5.19)	(−3.27)	(−6.54)	(−7.81)
All Stocks	−0.64	−0.04	0.04	0.01	0.09	−0.73	
	(−5.61)	(−0.58)	(0.76)	(0.29)	(2.12)	(−5.20)	
Panel B: 40% Smallest Stocks Omitted							
Most Overpriced	−1.44	−0.88	−0.56	−0.40	−0.39	−1.05	−0.74
(top 20%)	(−9.66)	(−6.12)	(−4.46)	(−3.11)	(−3.12)	(−5.42)	(−7.70)
Next 20%	−0.65	−0.25	−0.29	−0.21	−0.01	−0.63	−0.24
	(−4.15)	(−2.16)	(−2.89)	(−2.03)	(−0.15)	(−3.23)	(−4.16)
Next 20%	0.05	0.06	0.12	−0.12	0.03	0.02	−0.01
	(0.34)	(0.57)	(1.19)	(−1.25)	(0.34)	(0.11)	(−0.15)
Next 20%	0.03	0.15	0.14	0.18	0.20	−0.17	0.15
	(0.19)	(1.36)	(1.49)	(2.28)	(2.69)	(−0.90)	(3.46)
Most Underpriced	0.55	0.65	0.42	0.27	0.16	0.39	0.28
(Bottom 20%)	(3.21)	(5.30)	(4.18)	(3.25)	(2.17)	(2.00)	(5.61)
Most Overpriced −	−1.99	−1.52	−0.98	−0.67	−0.55	−1.44	−1.02
Most Underpriced	(−9.37)	(−7.72)	(−5.56)	(−4.19)	(−3.50)	(−6.00)	(−7.61)
All Stocks	−0.46	−0.05	0.04	0.03	0.09	−0.55	
	(−4.01)	(−0.67)	(0.77)	(0.79)	(2.00)	(−3.87)	

(Continued)

Table VII, panels A–B: market-cap thresholds

Table VII—*Continued*

	Highest IVOL	Next 20%	Next 20%	Next 20%	Lowest IVOL	Highest −Lowest	All Stocks
Panel C: 60% Smallest Stocks Omitted							
Most Overpriced	−1.25	−0.58	−0.51	−0.34	−0.23	−1.02	−0.64
(top 20%)	(−8.19)	(−3.97)	(−4.10)	(−2.74)	(−1.93)	(−5.07)	(−6.75)
Next 20%	−0.46	−0.25	−0.26	−0.20	−0.02	−0.44	−0.21
	(−3.23)	(−2.20)	(−2.63)	(−1.82)	(−0.24)	(−2.30)	(−3.76)
Next 20%	0.01	0.19	−0.03	−0.03	0.17	−0.16	0.04
	(0.09)	(1.90)	(−0.30)	(−0.27)	(1.73)	(−0.81)	(0.83)
Next 20%	0.22	0.05	0.11	0.19	0.21	0.01	0.15
	(1.38)	(0.47)	(1.13)	(2.21)	(2.36)	(0.06)	(3.08)
Most Underpriced	0.54	0.61	0.40	0.23	0.13	0.40	0.27
(Bottom 20%)	(3.15)	(5.29)	(4.14)	(2.63)	(1.74)	(2.08)	(5.17)
Most Overpriced −	−1.79	−1.19	−0.91	−0.57	−0.37	−1.42	−0.91
Most Underpriced	(−8.55)	(−6.08)	(−5.33)	(−3.39)	(−2.39)	(−6.10)	(−6.78)
All Stocks	−0.34	0.01	0.01	0.06	0.10	−0.45	
	(−3.11)	(0.15)	(0.14)	(1.20)	(2.18)	(−3.20)	
Panel D: 80% Smallest Stocks Omitted							
Most Overpriced	−0.88	−0.48	−0.59	−0.32	−0.12	−0.77	−0.56
(Top 20%)	(−5.55)	(−3.34)	(−4.51)	(−2.21)	(−0.98)	(−3.66)	(−5.89)
Next 20%	−0.30	−0.17	−0.12	−0.14	−0.03	−0.27	−0.17
	(−1.99)	(−1.54)	(−1.05)	(−1.36)	(−0.27)	(−1.27)	(−2.94)
Next 20%	0.02	0.06	0.06	0.12	0.17	−0.16	0.06
	(0.11)	(0.53)	(0.62)	(1.21)	(1.76)	(−0.80)	(1.30)
Next 20%	0.25	0.05	0.18	0.25	0.21	0.05	0.16
	(1.72)	(0.48)	(1.66)	(2.65)	(2.21)	(0.27)	(3.06)
Most Underpriced	0.51	0.43	0.44	0.28	0.03	0.47	0.26
(Bottom 20%)	(2.90)	(3.56)	(4.09)	(3.15)	(0.40)	(2.34)	(4.34)
Most Overpriced -	−1.39	−0.91	−1.03	−0.60	−0.16	−1.23	−0.82
Most Underpriced	(−6.23)	(−4.45)	(−5.59)	(−3.37)	(−0.99)	(−4.82)	(−5.86)
All Stocks	−0.19	−0.02	0.07	0.10	0.07	−0.26	
	(−1.80)	(−0.29)	(1.34)	(1.84)	(1.39)	(−1.88)	

Finally, we repeat the analysis in Table V under the same progressive elim

Table VII, panels C–D: market-cap thresholds

Table VIII

Idiosyncratic Volatility Effects and Investor Sentiment: Predictive Regressions under Various Thresholds for Market Capitalization

This table reports estimates of b in the regression

$$R_{i,t} = a + bS_{t-1} + cMKT_t + dSMB_t + eHML_t + u_t,$$

where $R_{i,t}$ is the excess percent return in month t and S_t is the level of the investor sentiment index of Baker and Wurgler (2006). The sort on IVOL is performed for stocks within a given range of over-underpricing, as determined by an average of the ranking percentiles produced by 11 anomaly variables. In each panel, the universe of stocks being sorted consists of all those with market capitalization exceeding a given percentile. Also reported are results based on sorting by IVOL within the entire universe. The sample period is from August 1965 to January 2011. All t-statistics are based on the heteroskedasticity-consistent standard errors of White (1980). In Panels A, B, C, and D, the smallest 20%, 40%, 60%, and 80% of the firms are omitted from the portfolio formation, respectively.

	Highest IVOL		Lowest IVOL		Highest − Lowest	
	b	t-Stat.	b	t-Stat.	b	t-Stat.
Panel A: 20% Smallest Stocks Omitted						
Most Overpriced (Top 20%)	−0.79	−3.82	0.00	0.04	−0.79	−3.54
Next 20%	−0.44	−2.83	0.10	1.10	−0.53	−2.80
Next 20%	−0.11	−0.84	0.31	3.38	−0.42	−2.41
Next 20%	−0.10	−0.65	0.08	0.95	−0.18	−0.97
Most Underpriced (Bottom 20%)	−0.07	−0.52	0.14	1.50	−0.20	−1.29
Most Overpriced − Most Underpriced	−0.72	−3.07	−0.13	−0.95	−0.59	−2.62
All Stocks	−0.46	−3.80	0.18	3.76	−0.64	−4.15
Panel B: 40% Smallest Stocks Omitted						
Most Overpriced (Top 20%)	−0.83	−4.03	−0.02	−0.24	−0.80	−3.56
Next 20%	−0.38	−2.31	0.15	1.60	−0.53	−2.58
Next 20%	−0.19	−1.48	0.31	3.13	−0.50	−2.80
Next 20%	0.01	0.11	0.12	1.53	−0.11	−0.61
Most Underpriced (Bottom 20%)	−0.03	−0.24	0.14	1.45	−0.17	−1.02
Most Overpriced − Most Underpriced	−0.79	−3.45	−0.16	−1.12	−0.63	−2.79
All Stocks	−0.43	−3.48	0.19	3.93	−0.62	−3.93
Panel C: 60% Smallest Stocks Omitted						
Most Overpriced (Top 20%)	−0.78	−3.86	0.04	0.39	−0.81	−3.77
Next 20%	−0.33	−2.14	0.11	1.23	−0.44	−2.25
Next 20%	−0.01	−0.09	0.27	2.64	−0.29	−1.51
Next 20%	0.03	0.21	0.14	1.73	−0.11	−0.70
Most Underpriced (Bottom 20%)	0.03	0.20	0.14	1.44	−0.11	−0.64
Most Overpriced − Most Underpriced	−0.80	−3.46	−0.10	−0.70	−0.70	−3.23
All Stocks	−0.35	−2.82	0.17	3.49	−0.52	−3.27
Panel D: 80% Smallest Stocks Omitted						
Most Overpriced (Top 20%)	−0.80	−3.87	0.11	0.94	−0.91	−3.93
Next 20%	−0.34	−2.29	0.18	1.75	−0.52	−2.63
Next 20%	0.00	0.01	0.29	2.82	−0.29	−1.52
Next 20%	0.02	0.16	0.16	1.76	−0.14	−0.85
Most Underpriced (Bottom 20%)	0.04	0.25	0.11	1.11	−0.07	−0.41
Most Overpriced − Most Underpriced	−0.84	−3.37	0.00	0.01	−0.84	−3.28
All Stocks	−0.31	−2.62	0.16	2.97	−0.47	−2.99

6 Short summary

- The paper explains the negative IVOL-return relation using arbitrage risk plus arbitrage asymmetry.
- IVOL makes arbitrage risky and allows mispricing to persist.
- Buying underpriced stocks is easier than shorting overpriced stocks.
- Therefore, high-IVOL overpriced stocks remain more overpriced than high-IVOL underpriced stocks remain underpriced.
- A composite of 11 anomalies identifies relatively overpriced and underpriced stocks.
- The IVOL-return relation is negative among overpriced stocks and positive among underpriced stocks.
- The negative effect is stronger, creating the overall IVOL puzzle.
- The negative effect is stronger among stocks with low institutional ownership and among smaller firms.
- High investor sentiment strengthens the negative IVOL effect among overpriced stocks.
- Low investor sentiment strengthens the positive IVOL effect among underpriced stocks.
- Results are robust to macro controls and to excluding smaller firms.

7 Conclusion

7.1 Core conclusion

Stambaugh, Yu, and Yuan show that the idiosyncratic volatility puzzle is best understood as a mispricing phenomenon constrained by asymmetric arbitrage. IVOL magnifies mispricing because it creates arbitrage risk, but the effect is stronger for overpricing because shorting is harder than buying.

7.2 Economic intuition

High IVOL is not itself the reason a stock earns low returns. The reason is that high IVOL protects mispricing from arbitrage. When a stock is overpriced, short sellers face borrowing constraints, margin risk, and limited capital. When a stock is underpriced, buyers face fewer constraints. Hence, high-IVOL overpricing persists more strongly than high-IVOL underpricing.

7.3 Takeaway

The paper reframes the IVOL puzzle as evidence about limits to arbitrage. To understand the IVOL-return relation, one must condition on the direction of mispricing and on the ease of arbitrage. Once these dimensions are added, the apparently puzzling negative IVOL effect becomes a prediction of arbitrage asymmetry.